

Fisher Information

Lecture/Lab 12

Physics 4730, Fall 2026



Ronald A. Fisher (1890 – 1962)

Introduction to Fisher Information

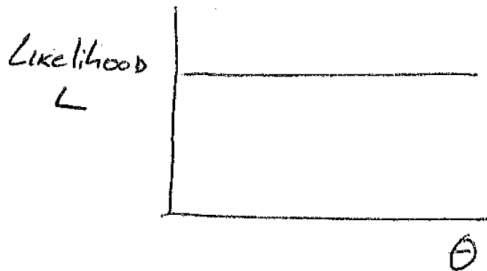
Q: How does randomness in data $\longrightarrow$ uncertainty in parameter estimation?

Q: How do we know if an estimator is the most efficient?

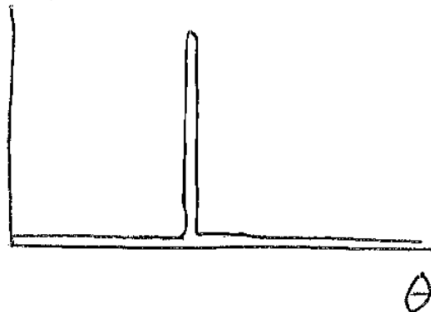
A: “Fisher Information” (Ronald Fisher, 1890 – 1962)

- ▶ Quantifies the information that RV x carries about $M(\theta)$
- ▶ The expected value of the observed information

Intuitive Introduction



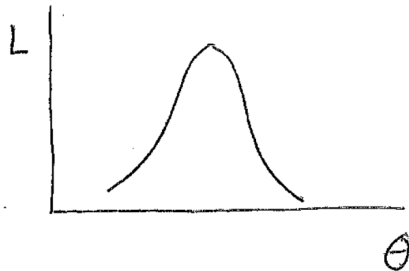
Measurement of x_i provides **no** information.



Measurement of x_i provides **all** information.

Intuitive Introduction

- ▶ The existence of a **slope** indicates that x_i provides information.
- ▶ The sign of the slope doesn't matter $\rightarrow$ square it!
- ▶ Take *average* $\langle \text{slope}^2 \rangle$ over the PDF ...



Measurement of x_i provides **some** information. **Most common scenario!**

Intuitive Introduction

- ▶ The existence of a **slope** indicates that x_i provides information.
- ▶ The sign of the slope doesn't matter $\rightarrow$ square it!
- ▶ Take *average* $\langle \text{slope}^2 \rangle$ over the PDF ...

$$\begin{aligned} I &= \int \left(\frac{\partial}{\partial \theta} \ln(f(x|\theta)) \right)^2 f(x|\theta) dx \\ &= E \left[\left(\frac{\partial}{\partial \theta} \ln(f(x|\theta)) \right)^2 \right] \end{aligned}$$

where E is the *Expectation Value*.

The information thus defined is **large** if, on average, $\ln f$ is sensitive to changes in θ .

Other Factoids about Fisher Information

1. $I(\theta) = \langle (\ln(f)')^2 \rangle = -\langle \ln(f)'' \rangle$ **Partial Integration**
2. $I_N(\theta) = N \times I_1(\theta)$ **Information for N measurements**
3. Cramer–Rao Inequality: For *unbiased estimators*, the variance of the estimator cannot be less than the *inverse* of Fisher Information.

$$\text{Var}(\theta) \geq \frac{1}{I_N(\theta)}$$

Example: Poisson Distribution

$$f(x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\ln f = x \ln \lambda - \lambda - \ln(x!)$$

$$\frac{\partial}{\partial \lambda} \ln f = \frac{x}{\lambda} - 1$$

$$\frac{\partial^2}{\partial \lambda^2} \ln f = -\frac{x}{\lambda^2}$$

$$I(\lambda) = \mathbb{E} \left[\left(\frac{x}{\lambda} - 1 \right)^2 \right] = \mathbb{E} \left[\frac{x}{\lambda^2} \right] = \frac{1}{\lambda}$$

Example: Normal Distribution

$$f(x|\theta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\theta)^2/2\sigma^2}$$

$$I(\theta) = \frac{1}{\sigma^2} \quad \text{you will show in lab!}$$

$$I_N(\theta) = \frac{N}{\sigma^2}$$

$$\text{Var}(\theta) \geq \frac{\sigma^2}{N} \quad \text{Cramer Rao}$$

$$\sigma_\theta \equiv \sqrt{\text{Var}(\theta)} \geq \frac{\sigma}{\sqrt{N}} \quad \text{error in the mean for normal distributions!}$$

Lab 12

→ Exercises to Demonstrate Fisher Information for the Normal Distribution.